

Deekshitha Devarasetty

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EDUCATION

Georgia Institute of Technology

B.S. in Computer Science - Systems & Architecture, Information Internetworks

Atlanta, GA

Graduation: May 2027

Related Coursework: Data Structures and Algorithms, Operating Systems, Database Systems, Artificial Intelligence, Computer Networks, Computer Organization, Linear Algebra, Internet Systems and Services, Compilers, Statistics

EXPERIENCE

Mastercard

June 2026 – August 2026

AI Engineering Intern

St. Louis, MO

- Built a predictive reliability engine to help SRE teams detect NATS infrastructure failures before customer impact
- Converted real-time Dynatrace telemetry into ML-ready signals for proactive incident detection and response
- Benchmarked anomaly detection and forecasting models to improve saturation-risk prediction accuracy
- Created explainable incident-risk outputs to help engineers identify degradation drivers and prioritize action
- Designed the system for future MCP/SRE bot integration, advancing automated incident response workflows
- Reduced false positives by 50% using an Isolation Forest + CUSUM ensemble, achieving 80.9% model agreement across 3,735 NATS CPU usage samples

Mastercard

June 2025 – August 2025

Fullstack Software Engineer Intern

St. Louis, MO

- Developed a pipeline that transforms NL prompts into functional Spring Boot RESTful APIs built by Gradle
- Reduced development time by 80% and deployed as a reusable internal tool
- Delivered a live technical demo to senior VPs and Directors, demonstrating end-to-end API creation, deployment to PCF, and integration with databases
- Coordinated with dev team in an Agile environment to accelerate feature delivery and enhance team productivity
- Created a Workday-integrated corporate card request prototype with workflow automation and API data exchange
- Collaborated with cross-functional teams to streamline request/approval pipelines and save time for employees

Automated Algorithm Design Lab

January 2024 – May 2025

Machine Learning Researcher

Atlanta, GA

- Built evolutionary algorithm evaluation pipelines with scikit-learn, DEAP, and MATLAB across controlled trials
- Ran ablation studies on selection, mutation, and evaluation strategies to improve generated solution performance
- Explored LLM-assisted program generation for genetic programming and automated algorithm design workflows

Bromberg Lab

May 2024 – June 2024

Data Analyst

Atlanta, GA

- Analyzed 12,000+ protein datasets with pandas and NumPy to uncover enzyme-family embedding patterns
- Evaluated EC-number clustering with scikit-learn by comparing enzyme labels and vector distances systematically
- Created publishable visual summaries to clarify enzyme similarity trends for biological downstream analysis results

PROJECTS

GPT-2 LLM | *Python, PyTorch, HuggingFace, HellaSwag, CUDA*

- Modernized GPT-2 architecture from scratch by transitioning from TensorFlow to PyTorch
- Improved OpenAI's GPT-2 LLM, enhancing a 124M parameter model with 12 layers, and a 50257 vocab size
- Leveraged FineWeb's 10BT sample to train the model and the HellaSwag evaluation to improve accuracy
- Surpassed OpenAI's benchmarks by achieving a lower validation loss and a higher learning accuracy
- Demonstrated advanced expertise in neural network architecture and multi-GPU model training

Foodify | *MongoDB, Express.js, React, Node.js*

- Developed a social food-tracking app where users share meals through photos and comments
- Utilized MongoDB as the database and React to ensure a dynamic and responsive user interface
- Leveraged Node.js to handle server-side logic and efficient routing for faster load times and improved scalability
- Employed Express.js to create REST APIs for data handling and communication between front end and database
- Enabled user authentication and authorization, allowing users to log in, create, and edit posts

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL, Kotlin, HTML/CSS

Frameworks: Spring Boot, FastAPI, React, Node.js, Express.js, PyTorch, HuggingFace

ML/AI: scikit-learn, LightGBM, SHAP, Prophet, Isolation Forest, CUSUM, DEAP, NumPy, pandas, MATLAB

Tools/Technologies: Git, Docker, Gradle, MongoDB, PostgreSQL, SQLite, Dynatrace, PCF, Insomnia